A Norm-Two Cohomology Anomaly at the Hénon Reflection Boundary

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Abstract

Primitive odd reflection points of the full Hénon horseshoe admit two natural sampling conventions. Marking the reflection center converges to a non-invariant boundary Bernoulli law η_J, whereas cyclic orbit averaging converges to the invariant maximal-entropy law. We prove that marked packet pressure is not invariant under symbolic coboundaries. Its anomaly is the functional

$$A_J(u) = \int (u - u \circ \sigma)\, d\eta_J,$$

whose dual total-variation norm is exactly two. Explicit locally constant functions v_r satisfy $A_J(v_r) = 2(1 - 2^{-r})$, so the extremal norm is approached by finite cylinders. By contrast, a coboundary sums to zero on every finite periodic orbit; consequently every uniformly orbit-averaged packet pressure is exactly gauge invariant before taking a limit. The result identifies orbit averaging as a necessary structural repair, while making no arithmetic trace or operator claim. An exact mutation-locked certificate accompanies the theorem.

1 Reflection-boundary sampling

The area-preserving Hénon map

$$H(x, y) = (6 - x^2 - y, x)$$

has a real chain-recurrent set conjugate to the full two-shift; its reversor is intertwined with sequence reversal. This follows from the Devaney–Nitecki horseshoe and Arai's hyperbolic plateau [2, 1]. We use the standard left shift $(\sigma\omega)_k = \omega_{k+1}$ on $\{0, 1\}^{\mathbb{Z}}$; basic symbolic cohomology terminology follows Lind and Marcus [3].

Let $(\xi_k)_{k\geq 0}$ be independent fair bits and define the marked reflection-boundary process

$$\omega_k = \omega_{-k} = \xi_k \qquad (k \geq 0). \tag{1}$$

Its law is denoted by η_J. Earlier packet equidistribution shows that marked primitive reflection points converge to η_J, while uniformly averaging their time origins converges to the fair two-sided Bernoulli law. The present paper asks whether the marked pressure is an intrinsic object under the usual potential relation

$$f' = f + u - u \circ \sigma. \tag{2}$$

For a continuous potential f, write the marked packet pressure as

$$P_J(f) = \frac{1}{2} \log 2 - \int f\, d\eta_J. \tag{3}$$

This is the limiting extensive pressure proved for the preceding packet family; the formula also defines its continuous extension to arbitrary symbolic potentials.

2 The boundary anomaly

Define

$$A_J(u) = \int (u - u \circ \sigma)\, d\eta_J. \tag{4}$$

Theorem 2.1 (Exact gauge law)**.** *For every continuous f, u,*

$$P_J(f + u - u \circ \sigma) = P_J(f) - A_J(u). \tag{5}$$

In particular, marked pressure descends to symbolic cohomology classes if and only if A_J vanishes identically.

Proof. Substitution of (2) into (3) gives (5). The last statement follows by testing every transfer function u. □

The functional is nonzero because η_J is not shift invariant. More is true. For a finite signed measure ν, use the dual norm

$$\|\nu\|_{\mathrm{TV}} = \sup_{\|u\|_\infty \le 1} \left| \int u\, d\nu \right|,$$

which equals two, rather than one, for the difference of mutually singular probability measures.

Theorem 2.2 (Norm-two anomaly)**.** *The measures η_J and $\sigma_* \eta_J$ are mutually singular, and*

$$\|A_J\| = \|\eta_J - \sigma_* \eta_J\|_{\mathrm{TV}} = 2. \tag{6}$$

Proof. The support of η_J consists of sequences fixed by reflection about the marked origin. The support of $\sigma_* \eta_J$ consists of sequences fixed by reflection about the adjacent center. A sequence satisfying both symmetries is invariant under their composition, a shift by two, and hence is period two. This finite set has probability zero under either iid half-word construction. The measures are therefore mutually singular. The difference of two mutually singular probability measures has dual total-variation norm two. Finally, $A_J(u) = \int u\, d(\eta_J - \sigma_* \eta_J)$. □

3 Finite-cylinder extremizers

The norm in theorem 2.2 is not merely an abstract measure-theoretic quantity. For $r \ge 1$, let

$$u_r(\omega) = \mathbf{1}_{\{\omega_{-k} = \omega_k \text{ for } 1 \le k \le r\}}, \qquad v_r = 2u_r - 1. \tag{7}$$

Proposition 3.1 (Local witnesses)**.** *For every $r \ge 1$,*

$$\int u_r\, d\eta_J = 1, \qquad \int u_r \circ \sigma\, d\eta_J = 2^{-r}, \qquad A_J(v_r) = 2(1 - 2^{-r}). \tag{8}$$

Thus $\|v_r\|_\infty = 1$ and $A_J(v_r) \to 2$.

Proof. The first equality is built into (1). After one shift, the r constraints become

$$\xi_{k-1} = \xi_{k+1}, \qquad 1 \le k \le r.$$

They leave precisely two free parity classes among the $r+2$ involved fair bits, so their probability is $2^2/2^{r+2} = 2^{-r}$. Affine centering then gives the third equality. □

This also gives a fully finite diagnosis: a radius-r cylinder already detects all but 2^{1-r} of the maximal anomaly.

4 Orbit averaging repairs the gauge

Let ω be periodic with period n. The coboundary identity telescopes:

$$\sum_{j=0}^{n-1}(u - u \circ \sigma)(\sigma^j \omega) = u(\omega) - u(\sigma^n \omega) = 0. \tag{9}$$

Corollary 4.1 (Finite-period gauge invariance)**.** *Every packet statistic formed by uniformly averaging a potential over each complete periodic orbit is unchanged by* (2)*, at every finite period and for every transfer function u.*

Proof. Apply (9) orbit by orbit before taking any packet sum or limit. □

The contrast is structural. Marked reflection sampling retains a boundary and therefore a norm-two gauge defect. Complete cyclic sampling removes the boundary exactly. This proves that the frozen coordinate-Mahler marked law is a valid coordinate-dependent statistic, but not a canonical Livšic invariant. It also isolates the next theorem: among normalized linear samplers on an n-cycle, is uniform cyclic averaging the unique sampler that annihilates every coboundary?

5 Executable certificate and claim boundary

The accompanying exact-arithmetic programs verify (8) for $1 \le r \le 12$, mutual-singularity support intersections on finite windows, and telescoping on every binary word through period 10. Two independent implementations agree, seven unit tests pass in normal and optimized modes, and 19 hostile mutations of signs, shifts, normalizations, and endpoint claims are rejected. No floating-point evidence enters the theorem.

This paper proves a symbolic pressure obstruction and its exact repair. It does not produce rational-prime labels, von Mangoldt amplitudes, a Fredholm determinant, a self-adjoint operator, or a Hilbert–Pólya correspondence. Route A therefore remains exploratory and Route B is not authorized.

References

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